

Characteristic Classes

Book Notes of *Characteristic Classes*

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Contents

1	Multiplicative sequences	2
1.1	Multiplicative Sequences	2
1.2	Characteristic Power Series	4
1.3	K-genus	6
1.4	Signature	7
1.5	Exercise	11

1 Multiplicative sequences

We start with polynomial ring $\mathbb{Q}[\beta_1, \beta_2, \dots]$ in countably many variables β_i over the rational number field $\mathbb{Q}$. Let $A = \mathbb{Q}[a_1, a_2, \dots] = \text{Sym}[\beta_1, \beta_2, \dots]$ be the subring of symmetric polynomials, where a_i is the i -th elementary symmetric polynomial in $\beta_1, \beta_2, \dots$. A has a grading structure by setting $\deg a_i = i$.

$$A = \bigoplus_{n=0}^{\infty} A^n, \quad A^n = \{\text{homogeneous polynomials of degree } n\}.$$

We set

$$a = 1 + a_1 t + a_2 t^2 + \dots = \prod_{i=1}^{\infty} (1 + \beta_i t) \in A[[t]],$$

For any two

$$A = \mathbb{Q}[a_1, a_2, \dots] \subset \mathbb{Q}[\beta_1, \beta_2, \dots], \quad B = \mathbb{Q}[b_1, b_2, \dots] \subset \mathbb{Q}[\beta'_1, \beta'_2, \dots]$$

and

$$a = 1 + a_1 t + a_2 t^2 + \dots, \quad b = 1 + b_1 t + b_2 t^2 + \dots$$

their product is

$$\begin{aligned} ab &= (1 + a_1 t + a_2 t^2 + \dots)(1 + b_1 t + b_2 t^2 + \dots) \\ &= 1 + (a_1 + b_1)t + (a_2 + a_1 b_1 + b_2)t^2 + \dots \end{aligned}$$

Remark. If we take $t = 1$ and $a_i = c_i(\omega)$, where ω is a complex n -plane bundle, then

$$a = 1 + c_1(\omega) + c_2(\omega) + \dots + c_n(\omega) = c(\omega)$$

1.1 Multiplicative Sequences

Let $\{K_n\}$ be a sequence of polynomials:

$$K_0 = 1, \quad K_1 = K_1(x_1), \quad K_2 = K_2(x_1, x_2), \quad K_3 = K_3(x_1, x_2, x_3), \quad \dots,$$

satisfying that each

$$K_n(a_1, \dots, a_n) \in A^n$$

is a homogeneous polynomial of degree n .

For any

$$a = 1 + a_1t + a_2t^2 + \cdots ,$$

we define

$$K(a) = 1 + K_1(a_1)t + K_2(a_1, a_2)t^2 + K_3(a_1, a_2, a_3)t^3 + \cdots .$$

We say that $\{K_n\}$ is a **multiplicative sequence** (or m-sequence) if for any two such a and b as above, we have

$$K(ab) = K(a)K(b)$$

which forces that

$$\begin{aligned} K_1(a_1 + b_1) &= K_1(a_1) + K_1(b_1), \\ K_2(a_1 + b_1, a_2 + a_1b_1 + b_2) &= K_2(a_1, a_2) + K_1(a_1)K_1(b_1) + K_2(b_1, b_2), \\ &\dots\dots \end{aligned}$$

Example 1.1. Set $K_n(x_1, \dots, x_n) = \lambda^n x_n$, where λ is a constant. Then we have

$$K(1 + a_1t + a_2t^2 + \cdots) = 1 + \lambda a_1t + \lambda^2 a_2t^2 + \cdots$$

Especially, if $\lambda = 1$, then

$$K(a) = a.$$

If $\lambda = -1$, then

$$K(a) = 1 - a_1t + a_2t^2 - a_3t^3 + \cdots .$$

If in addition we take $t = 1$, $a_i = c_i(\omega)$, then we get

$$K(c(\omega)) = 1 - c_1(\omega) + c_2(\omega) - \cdots + (-1)^n c_n(\omega) = c(\bar{\omega}).$$

Example 1.2. The identity $K(a) = a^{-1}$ defines a multiplicative sequence, where

$$\begin{aligned} K_0 &= 1, \\ K_1(x_1) &= -x_1, \\ K_2(x_1, x_2) &= x_1^2 - x_2, \\ K_3(x_1, x_2, x_3) &= -x_1^3 + 2x_1x_2 - x_3, \\ K_4(x_1, x_2, x_3, x_4) &= x_1^4 - 3x_1^2x_2 + 2x_1x_3 + x_2^2 - x_4, \\ &\dots\dots \end{aligned}$$

If we take $t = 1$, $a_i = c_i(\omega)$ and $\eta \oplus \omega = \varepsilon^{m+n}$, then we get

$$K(c(\omega)) = c(\omega)^{-1} = c(\eta).$$

Example 1.3. The polynomials

$$\begin{aligned} K_{2n+1} &= 0, \\ K_{2n} &= x_n^2 - 2x_{n-1}x_{n+1} + \cdots + (-1)^{n-1}2x_1x_{2n-1} + (-1)^n2nx_{2n} \end{aligned}$$

form an m-sequence. If we take $t = 1$, $a_i = c_i(\omega)$, then we get the total Pontryagin class

$$K(c(\omega)) = 1 + p_1(\omega_{\mathbb{R}}) + p_2(\omega_{\mathbb{R}}) + \cdots = p(\omega_{\mathbb{R}}).$$

1.2 Characteristic Power Series

Given a multiplicative sequence $\{K_n\}$, we can define its **characteristic power series** $f(t)$ by

$$K(1+t) = f(t) = 1 + \lambda_1 t + \lambda_2 t^2 + \lambda_3 t^3 + \cdots,$$

where $\lambda_i = K_i(1, 0, \dots, 0)$. The following lemma shows that there is a one-to-one correspondence between multiplicative sequences and power series.

Lemma 1.4 (Hirzebruch). *Let $f(t) = 1 + \lambda_1 t + \lambda_2 t^2 + \lambda_3 t^3 + \cdots$ be a power series in $\mathbb{Q}[[t]]$ with constant term 1. Then there **exists** a **unique** multiplicative sequence $\{K_n\}$ whose characteristic power series is $f(t)$.*

Proof. Uniqueness: Suppose that $\{K_n\}$ is a multiplicative sequence with characteristic power series $f(t)$. Then

$$K(1 + a_1 t + a_2 t^2 + \cdots) = K\left(\prod_{i=1}^{\infty} (1 + \beta_i t)\right) = \prod_{i=1}^{\infty} K(1 + \beta_i t) = \prod_{i=1}^{\infty} f(\beta_i t).$$

take $\beta_{n+1} = \beta_{n+2} = \cdots = 0$, we have

$$K(1 + a_1 t + a_2 t^2 + \cdots + a_n t^n) = \prod_{i=1}^n f(\beta_i t).$$

by comparing the coefficient of t^n on both sides,

$$K_n(a_1, a_2, \dots, a_n) = \text{the coefficient of } t^n \text{ in } \prod_{i=1}^n f(\beta_i t).$$

Since the right-hand side is a symmetric polynomial in $\beta_1, \dots, \beta_n$, it can be expressed as a polynomial in $a_1, \dots, a_n$. And $a_1, \dots, a_n$ are algebraically independent, so K_n is uniquely determined.

Existence: Define K_n by the above formula, then it is easy to check that $\{K_n\}$ is a multiplicative sequence whose characteristic power series is $f(t)$. $\square$

Remark. Given a power series

$$f(t) = 1 + \lambda_1 t + \lambda_2 t^2 + \lambda_3 t^3 + \cdots ,$$

the corresponding multiplicative sequence $\{K_n\}$ can be computed by the following: Denote $a_i^{(n)} = a_i(\beta_1, \dots, \beta_n, 0, 0, \dots)$, then

$$\begin{aligned} K_1(a_1^{(1)}) &= \text{the coefficient of } t \text{ in } f(\beta_1 t) \\ &= \lambda_1 \beta_1 = \lambda_1 a_1^{(1)} \\ K_2(a_1^{(2)}, a_2^{(2)}) &= \text{the coefficient of } t^2 \text{ in } f(\beta_1 t) f(\beta_2 t) \\ &= \lambda_1^2 \beta_1 \beta_2 + \lambda_2 (\beta_1^2 + \beta_2^2), \\ &= \lambda_1^2 a_2^{(2)} + \lambda_2 ((a_1^{(2)})^2 - 2a_2^{(2)}) \\ K_3(a_1^{(3)}, a_2^{(3)}, a_3^{(3)}) &= \text{the coefficient of } t^3 \text{ in } f(\beta_1 t) f(\beta_2 t) f(\beta_3 t) \\ &= \lambda_1^3 \beta_1 \beta_2 \beta_3 + \lambda_1 \lambda_2 (\beta_1 \beta_2^2 + \beta_2 \beta_3^2 + \beta_3 \beta_1^2) + \lambda_3 (\beta_1^3 + \beta_2^3 + \beta_3^3) \\ &= \lambda_1^3 a_3^{(3)} + \lambda_1 \lambda_2 (a_1^{(3)} a_2^{(3)} - 3a_3^{(3)}) + \lambda_3 ((a_1^{(3)})^3 - 3a_1^{(3)} a_2^{(3)} + 3a_3^{(3)}) \\ &\dots\dots \end{aligned}$$

Example 1.5. The characteristic power series of the multiplicative sequence in Example 1.1 is $f(t) = 1 + \lambda t$.

Example 1.6. The characteristic power series of the multiplicative sequence in Example 1.2 is

$$f(t) = \frac{1}{1+t} = 1 - t + t^2 - t^3 + t^4 - \cdots .$$

We can check the first few polynomials:

$$\begin{aligned} K_0 &= 1, \\ K_1(x_1) &= \lambda_1 x_1 = -x_1, \\ K_2(x_1, x_2) &= \lambda_1^2 x_2 + \lambda_2 (x_1^2 - 2x_2) = x_1^2 - x_2, \\ K_3(x_1, x_2, x_3) &= \lambda_1^3 x_3 + \lambda_1 \lambda_2 (x_1 x_2 - 3x_3) + \lambda_3 (x_1^3 - 3x_1 x_2 + 3x_3) = -x_1^3 + 2x_1 x_2 - x_3, \\ &\dots\dots \end{aligned}$$

Example 1.7. The characteristic power series of the multiplicative sequence in Example 1.3 is $f(t) = 1 + t^2$. We can check the first few polynomials:

$$\begin{aligned} K_0 &= 1, \\ K_1(x_1) &= \lambda_1 x_1 = 0, \end{aligned}$$

$$\begin{aligned}
K_2(x_1, x_2) &= \lambda_2 x_2 = x_2, \\
K_3(x_1, x_2, x_3) &= \lambda_1^3 x_3 + \lambda_1 \lambda_2 (x_1 x_2 - 3x_3) + \lambda_3 (x_1^3 - 3x_1 x_2 + 3x_3) = 0, \\
K_4(x_1, x_2, x_3, x_4) &= \lambda_2^2 (x_2^2 - 2x_1 x_3 + 2x_4) = x_2^2 - 2x_1 x_3 + 2x_4, \\
&\dots\dots
\end{aligned}$$

1.3 K-genus

Definition 1.8. Given a multiplicative sequence $\{K_n\}$. For any smooth, closed, oriented manifold M^m , we can define its **K-genus** by

$$\begin{aligned}
K[M] &= 0, \quad \text{if } 4 \nmid m, \\
K[M] &= \langle K_n(p_1(\tau_M), \dots, p_n(\tau_M)), \mu_M \rangle, \quad \text{if } m = 4n,
\end{aligned}$$

Thus $K[M]$ is a certain rational linear combination of Pontryagin numbers of M .

Remark. A K-genus can be written briefly as

$$K[M] = \langle K(p(\tau_M)), \mu_M \rangle,$$

Lemma 1.9. A K-genus gives rise to a ring homomorphism

$$\Omega_*^{SO} \rightarrow \mathbb{Q}, \quad [M] \mapsto K[M]$$

from the oriented cobordism ring to the rational number field.

Equivalently, this correspondence gives an algebra homomorphism from $\Omega_*^{SO} \otimes \mathbb{Q}$ to $\mathbb{Q}$. Thus defines an element in the dual space $\text{Hom}_{\mathbb{Q}}(\Omega_*^{SO} \otimes \mathbb{Q}, \mathbb{Q})$.

Proof. The correspondence is well-defined since Pontryagin numbers are cobordism invariants. It is clear that K is additive under disjoint union.

It suffices to show that for any two M and M' , we have

$$K[M \times M'] = K[M]K[M'].$$

If either $\dim M$ or $\dim M'$ is not divisible by 4, then both sides are zero. Now suppose that $\dim M = 4m$ and $\dim M' = 4n$. Then

$$\begin{aligned}
K[M \times M'] &= \langle K(p(\tau_{M \times M'})), \mu_{M \times M'} \rangle \\
&= \langle K(p(\tau_M) \times p(\tau_{M'})), \mu_M \times \mu_{M'} \rangle \\
&= \langle K(p(\tau_M)) \times K(p(\tau_{M'})), \mu_M \times \mu_{M'} \rangle \\
&= \langle K(p(\tau_M)), \mu_M \rangle \cdot \langle K(p(\tau_{M'})), \mu_{M'} \rangle \\
&= K[M]K[M'].
\end{aligned}$$

□

Remark. The multiplicative sequence and K-genus give a systematic machinery to produce various invariants of manifolds.

1.4 Signature

Definition 1.10. The **signature** $\sigma(M)$ of a smooth, closed, oriented $4k$ -manifold M is defined to be the signature of the symmetric bilinear form

$$Q_M : H^{2k}(M; \mathbb{Q}) \times H^{2k}(M; \mathbb{Q}) \rightarrow \mathbb{Q}, \quad Q_M(a, b) = \langle a \cup b, \mu_M \rangle.$$

For M with $4 \nmid \dim M$, we set $\sigma(M) = 0$.

To be precise, if we choose a basis $\{a_1, \dots, a_r\}$ of $H^{2k}(M; \mathbb{Q})$, so that the matrix of Q_M

$$Q_{ij} = \langle a_i \cup a_j, \mu_M \rangle,$$

is diagonal. Then the signature $\sigma(M)$ is defined to be the number of positive entries minus the number of negative entries on the diagonal. Notes that Q_M is non-degenerate by Poincaré duality, so the matrix (Q_{ij}) is invertible.

Lemma 1.11 (Thom). *The signature $\sigma(M)$ gives rise to a ring homomorphism*

$$\Omega_*^{SO} \rightarrow \mathbb{Z}, \quad [M] \mapsto \sigma(M).$$

Equivalently, this is to say the signature function satisfies:

1. $\sigma(M \sqcup M') = \sigma(M) + \sigma(M')$,
2. $\sigma(M \times M') = \sigma(M)\sigma(M')$.
3. *If $M = \partial W$ for some oriented manifold W , then $\sigma(M) = 0$.*

Proof. The first property follows by

$$H^{2k}(M \sqcup M'; \mathbb{Q}) \cong H^{2k}(M; \mathbb{Q}) \oplus H^{2k}(M'; \mathbb{Q}).$$

The second property follows by the Künneth formula

$$H^*(M \times M'; \mathbb{Q}) \cong H^*(M; \mathbb{Q}) \otimes H^*(M'; \mathbb{Q}).$$

For the third property, we need to show that if $M = \partial W$, then $\sigma(M) = 0$. By the long exact sequence of the pair (W, M) and Poincaré-Lefschetz duality, we have the following commutative diagram:

$$\begin{array}{ccccc} H^{2k}(W, M; \mathbb{Q}) & \xrightarrow{i^*} & H^{2k}(W; \mathbb{Q}) & \xrightarrow{j^*} & H^{2k}(M; \mathbb{Q}) \\ \downarrow \cong & & \downarrow \cong & & \downarrow \cong \\ H_{4k-2k}(W; \mathbb{Q}) & \xrightarrow{j_*} & H_{4k-2k}(W, M; \mathbb{Q}) & \xrightarrow{i_*} & H_{4k-2k-1}(M; \mathbb{Q}) \end{array}$$

Thus we have an exact sequence

$$H^{2k}(W, M; \mathbb{Q}) \xrightarrow{i^*} H^{2k}(W; \mathbb{Q}) \xrightarrow{j^*} H^{2k}(M; \mathbb{Q}) \xrightarrow{\delta} H^{2k+1}(W, M; \mathbb{Q}).$$

Note that for any $a, b \in H^{2k}(M; \mathbb{Q})$, if either a or b is in $\text{Im } j^*$, then

$$Q_M(a, b) = \langle a \cup b, \mu_M \rangle = 0,$$

since $j_*(\mu_M) = 0$. Thus Q_M descends to a well-defined bilinear form on

$$V = H^{2k}(M; \mathbb{Q}) / \text{Im } j^*.$$

Moreover, by the exactness of the above sequence, we have an isomorphism

$$V \cong \text{Im } \delta \subset H^{2k+1}(W, M; \mathbb{Q}).$$

Since $\dim H^{2k+1}(W, M; \mathbb{Q})$ is even, $\dim V$ is also even. On the other hand, the bilinear form Q_M on V is skew-symmetric, i.e. $Q_M([a], [b]) = -Q_M([b], [a])$ for any $[a], [b] \in V$. Thus the signature of Q_M on V is zero. Since Q_M vanishes on $\text{Im } j^*$, we have

$$\sigma(M) = \sigma(Q_M) = \sigma(Q_M|_V) = 0.$$

□

Remark. This lemma shows that the signature determine an element in the dual space $\text{Hom}_{\mathbb{Q}}(\Omega_*^{SO} \otimes \mathbb{Q}, \mathbb{Q})$ which takes values in $\mathbb{Z}$ for each oriented manifold.

Since

$$\Omega_*^{SO} \otimes \mathbb{Q} \cong \mathbb{Q}[\mathbb{C}P^2, \mathbb{C}P^4, \dots]$$

is a polynomial ring generated by the complex projective spaces, and the dual space

$$\text{Hom}_{\mathbb{Q}}(\Omega_*^{SO} \otimes \mathbb{Q}, \mathbb{Q}) \cong \text{span}_{\mathbb{Q}}\{p_I\}_I$$

is spanned by the Pontryagin numbers. So the signature function can be written as a linear combination of Pontryagin numbers.

Theorem 1.12 (Signature Theorem, Hirzebruch). *The signature $\sigma(M)$ of a smooth, closed, oriented $4k$ -manifold M is equal to the L -genus of M , i.e.*

$$\sigma(M) = \langle L_k(p_1(\tau_M), \dots, p_k(\tau_M)), \mu_M \rangle,$$

where L_k is the multiplicative sequence with characteristic power series

$$f(t) = \frac{\sqrt{t}}{\tanh \sqrt{t}} = 1 + \frac{t}{3} - \frac{t^2}{45} + \frac{2t^3}{945} - \dots + (-1)^{k-1} \frac{2^{2k} B_{2k}}{(2k)!} t^k + \dots,$$

Here B_{2k} are the Bernoulli numbers defined by the generating function

$$\frac{t}{e^t - 1} = 1 - \frac{t}{2} + \sum_{k=1}^{\infty} B_{2k} \frac{t^{2k}}{(2k)!}.$$

The first few Bernoulli numbers are

$$B_0 = 1, \quad B_1 = \frac{1}{6}, \quad B_2 = \frac{1}{30}, \quad B_3 = \frac{1}{42}, \quad B_4 = \frac{1}{30}, \quad B_5 = \frac{5}{66}, \quad \dots$$

The first few L -polynomials are

$$\begin{aligned} L_0 &= 1, \\ L_1 &= \frac{1}{3}p_1, \\ L_2 &= \frac{1}{45}(7p_2 - p_1^2), \\ L_3 &= \frac{1}{945}(62p_3 - 13p_1p_2 + 2p_1^3), \\ L_4 &= \frac{1}{14175}(381p_4 - 71p_1p_3 - 19p_2^2 + 22p_1^2p_2 - 3p_1^4), \\ &\dots\dots \end{aligned}$$

Here we compute the first three L -polynomials as follows:

$$\begin{aligned} L_1(p_1) &= \lambda_1 p_1 = \frac{1}{3}p_1, \\ L_2(p_1, p_2) &= \lambda_1^2 p_2 + \lambda_2(p_1^2 - 2p_2) = \frac{1}{9}p_2 - \frac{1}{45}(p_1^2 - 2p_2) = \frac{7}{45}p_2 - \frac{1}{45}p_1^2, \\ L_3(p_1, p_2, p_3) &= \lambda_1^3 p_3 + \lambda_1 \lambda_2(p_1 p_2 - 3p_3) + \lambda_3(p_1^3 - 3p_1 p_2 + 3p_3) \\ &= \frac{1}{27}p_3 - \frac{1}{135}(p_1 p_2 - 3p_3) + \frac{2}{945}(p_1^3 - 3p_1 p_2 + 3p_3) \\ &= \frac{62}{945}p_3 - \frac{13}{945}p_1 p_2 + \frac{2}{945}p_1^3. \end{aligned}$$

Proof of the Signature Theorem. Since the correspondences

$$\begin{aligned} \Omega_*^{SO} \otimes \mathbb{Q} &\rightarrow \mathbb{Q}, & [M] &\mapsto \sigma(M) \\ \Omega_*^{SO} \otimes \mathbb{Q} &\rightarrow \mathbb{Q}, & [M] &\mapsto L[M] \end{aligned}$$

are both ring homomorphisms, it suffices to check that they agree on the polynomial generators $\mathbb{C}P^2, \mathbb{C}P^4, \dots$ of $\Omega_*^{SO} \otimes \mathbb{Q}$.

Step 1: Compute $\sigma(\mathbb{C}P^{2k})$. Since $H^k(\mathbb{C}P^{2k}; \mathbb{Q})$ is generated by a^k with

$$\langle a^k \smile, \mu_{\mathbb{C}P^{2k}} \rangle = 1,$$

Thus $\sigma(\mathbb{C}P^{2k}) = 1$.

Step 2: Compute $L[\mathbb{C}P^{2k}]$. The total Pontryagin class of $\mathbb{C}P^{2k}$ is equal to

$$p = p(\tau_{\mathbb{C}P^{2k}}) = (1 + a^2)^{2k+1},$$

Since the characteristic power series of L is

$$f(t) = \frac{\sqrt{t}}{\tanh \sqrt{t}}$$

We have

$$\begin{aligned} L(p) &= [f(a^2)]^{2k+1} = \left(\frac{a}{\tanh a} \right)^{2k+1} \\ &= \left(1 - \frac{a^2}{3} + \frac{2a^4}{15} - \frac{17a^6}{315} + \cdots \right)^{2k+1}. \end{aligned}$$

Thus

$$L[\mathbb{C}P^{2k}] = \langle L(p), \mu_{\mathbb{C}P^{2k}} \rangle = \text{the coefficient of } a^{2k} \text{ in } \left(\frac{a}{\tanh a} \right)^{2k+1}.$$

In order to compute this coefficient, we use the residue theorem in complex analysis: let

$$g(z) = \left(\frac{z}{\tanh z} \right)^{2k+1}$$

be a complex function. Then the coefficient of z^{2k} in $g(z)$ is equal to

$$\begin{aligned} \frac{1}{2\pi i} \int_{|z|=\varepsilon} \frac{g(z)}{z^{2k+1}} dz &= \frac{1}{2\pi i} \int_{|z|=\varepsilon} \frac{dz}{(\tanh z)^{2k+1}} \\ &= \frac{1}{2\pi i} \int_{|u|=\varepsilon'} \frac{du}{(1-u^2)u^{2k+1}} \quad (u = \tanh z) \\ &= \frac{1}{2\pi i} \int_{|u|=\varepsilon'} \frac{1+u^2+u^4+\cdots}{u^{2k+1}} du \\ &= 1, \end{aligned}$$

where ε and ε' are sufficiently small. Thus we have $L[\mathbb{C}P^{2k}] = 1$. $\square$

A more direct proof of the signature theorem is given by Atiyah and Singer in their famous index theorem for elliptic operators.

Corollary 1.13. *The L -genus $L[M]$ of any smooth, closed, oriented $4k$ -manifold M is an integer.*

Proof. Since $L[M] = \sigma(M) \in \mathbb{Z}$. □

It follows that the Pontryagin numbers of any smooth, closed, oriented $4k$ -manifold satisfy certain integrality condition. For example,

$$\begin{aligned} L_1 &= \frac{1}{3}p_1 \Rightarrow p_1[\tau_M] \equiv 0 \pmod{3}, \\ L_2 &= \frac{1}{45}(7p_2 - p_1^2) \Rightarrow 7p_2[\tau_M] - p_1^2[\tau_M] \equiv 0 \pmod{45}. \end{aligned}$$

Corollary 1.14. *The L -genus $L[M]$ is an oriented homotopy invariant of M .*

Proof. Since $\sigma(M)$ is invariant under orientation-preserving homotopy equivalences. □

1.5 Exercise

Problem (19-A). Let T_n be the multiplicative sequence of polynomials belonging to the power series $f(t) = \frac{t}{1-e^{-t}}$. Then the **Todd genus** $T[M]$ of a complex n -dimensional manifold is defined to be the characteristic number $\langle T_n(c_1(\tau_M), \dots, c_n(\tau_M)), \mu_M \rangle$. Prove that $T[\mathbb{C}P^n] = +1$ and prove that T_n is the only multiplicative sequence with this property.